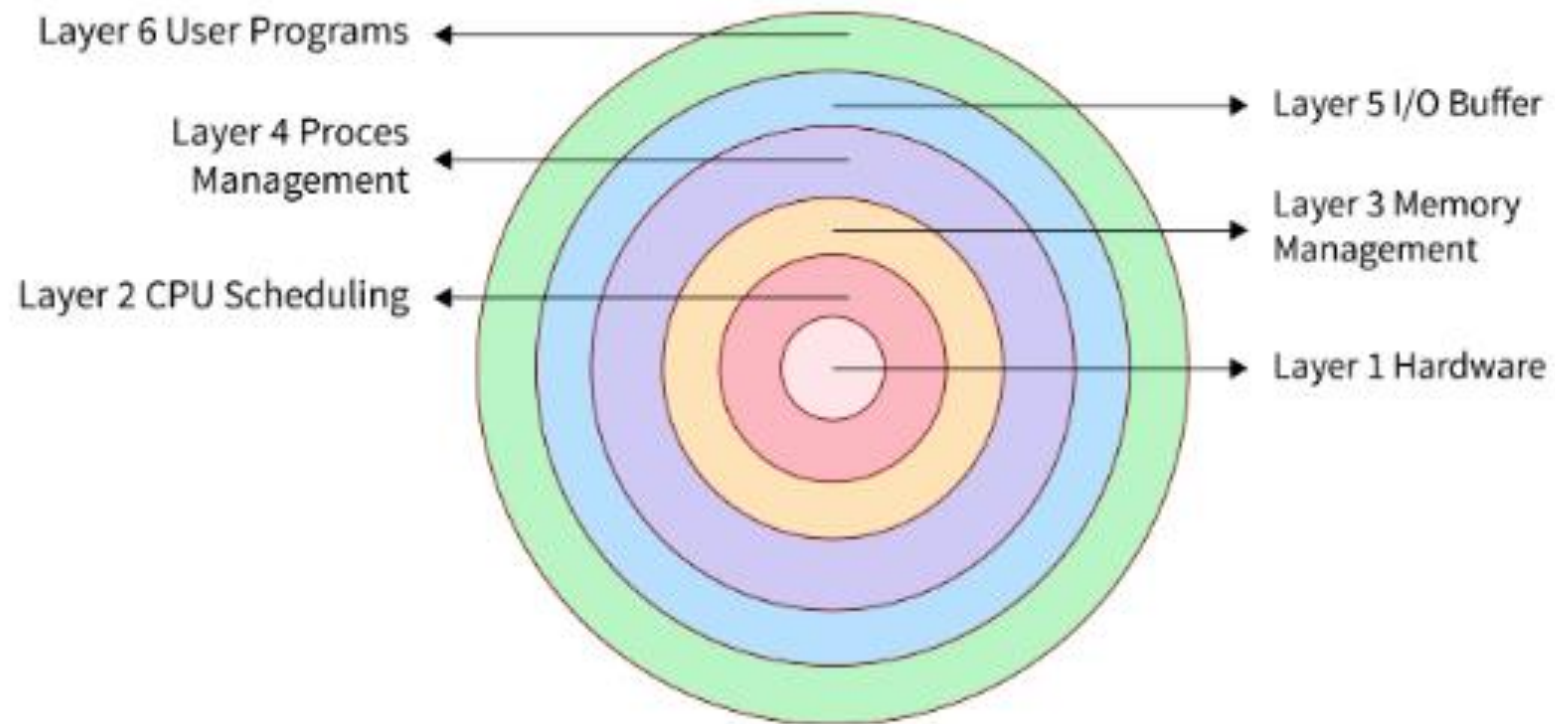


Layered Architecture (Windows Xp and 7)

- One way to achieve modularity in the operating system is the layered approach. In this, the **bottom layer is the hardware** and the **topmost layer is the user interface**.
- Windows uses a **modular layered architecture**.
- Each layer provides services to the layer above it and relies on the layer below it.
- This design improves **portability, stability, and security**.

Layered Architecture (Windows Xp and 7)



Advantages

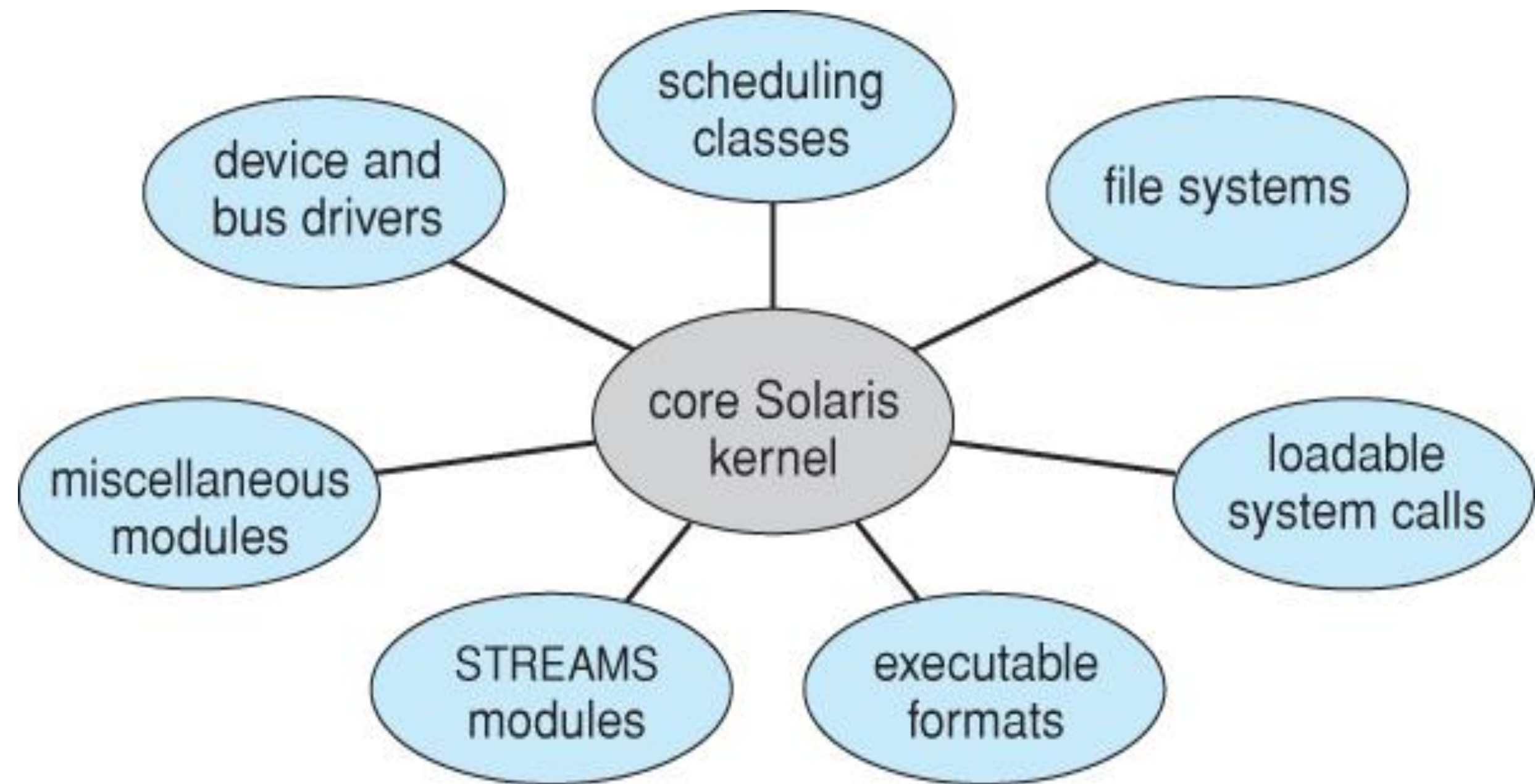
- **High Customizable** - Being layered, each layer implementation can be customized easily. A new functionality can be added without impacting other modules as well.
- **Verifiable** - Being modular, each layer can be verified and debugged easily.

Disadvantages

- **Complex designing** - Each layer is to planned carefully as each layer communicates with lower layer only and a good design process is required to create a layered operating system.

Modular Architecture

- Modular architecture operating system works on the similar principle as a monolith but with better design.
- A central kernel is responsible for all major operations of operating system.
- This kernel has set of core functionality and other services are loaded as modules dynamically to the kernel at boot time or at runtime.
- Sun Solaris OS is one of the example of Modular structured operating system.



Single User Operating System (SUS)

- A Single User Operating System (SUS) is an operating system **designed to be used by one user at a time**, and it does not support **multiple users accessing the system simultaneously**.
- The resources of the system (such as CPU, memory, and I/O devices) are **dedicated to a single user**, and only that user can run processes or interact with the system during its operation.

Features of Single User Operating System

- **Single User Access:** Only one user can use the system at a time.
- **Simple Design:** Easier to develop and manage because it does not handle multiple users simultaneously.
- **Dedicated Resource Allocation:** CPU, memory, and storage resources are primarily allocated to a single user.
- **Supports Multitasking:** Can run multiple applications for one user at the same time.
- **Less Complex Security:** Requires fewer authentication and access control mechanisms.
- **User-Friendly Environment:** Easy to learn and operate for individual users.
- **Efficient Performance:** Resources are not shared among multiple users, improving responsiveness.
- **Suitable for Personal Computing:** Commonly used on personal computers, laptops, and home systems.

Benefits of Single User Operating System

- **Simple to Use and Manage**
- **Optimized Performance for One User**
- **Lower System Complexity**
- **Reduced Administrative Requirements**
- **Cost-Effective for Personal Use**
- **Quick Response Time**
- **Easy Installation and Maintenance**
- **Ideal for Individual Productivity Tasks**

Limitations of Single User Operating System

- **No Multi-User Support:** Multiple users cannot access the system simultaneously.
- **Limited Scalability:** Not suitable for large organizations or shared environments.
- **Restricted Resource Sharing:** Designed for individual rather than collaborative use.
- **Less Suitable for Networked Environments**
- **Security Risks in Shared Usage:** Data separation may be limited if multiple people use the same machine.
- **Not Ideal for Enterprise Applications**
- **Limited Centralized Management**
- **Lower Resource Utilization Compared to Multi-User Systems**

Multi-User Operating System

An operating system is software that acts as an interface between the user and the computer hardware which does multiple functions such as **memory management; file management and processor management.**

The operating system should have to meet the requirements of all its users in a balanced way so that if any problem would arise with a user, it does not affect any other user in the chain.

Multi-User Operating System

1. In a multiuser operating system, **multiple numbers of users can access different resources of a computer at the same time.**
2. The access is provided using a **network** that consists of various personal computers attached to a **mainframe** computer system.
3. A multi-user operating system allows the permission of **multiple users for accessing a single machine at a time.**
4. The various personal computers can send and receive information to the mainframe computer system. Thus, the **mainframe computer acts as the server** and other personal computers **act as clients for that server.**

Types of Multi-user Operating Systems:

- The multi-user operating systems is of the following types:

1. Distributed System

2. Time sliced system

3. Multiprocessor system

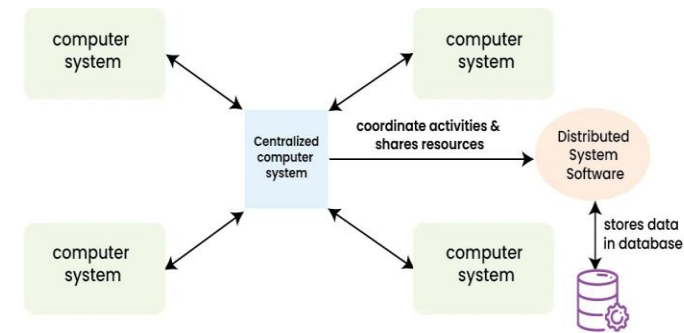
Distributed system:

Distributed operating systems manage a network of separate computers as a unified system. They provide a unified interface to users and applications while balancing the load and resources across multiple machines. These systems improve scalability and reliability.

Examples:

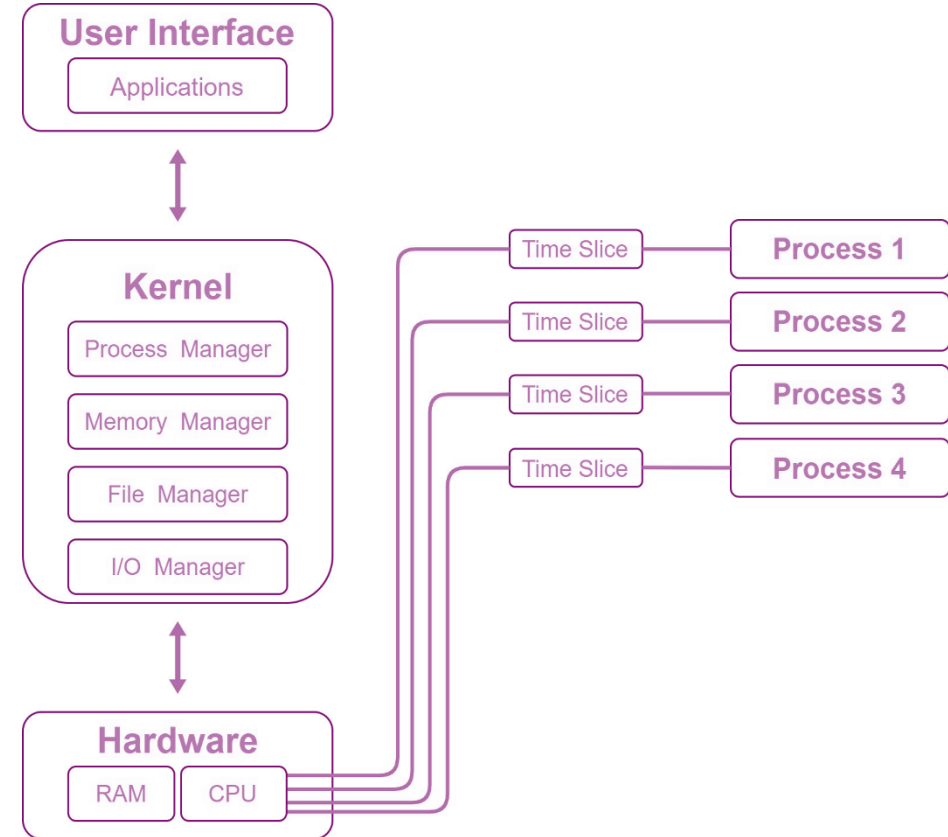
Apache Hadoop - Hadoop is a distributed operating system for processing and storing large datasets across a cluster of computers. It handles data distribution and fault tolerance across a network of machines.

1. Electronic banking,
2. Mobile apps (Facebook, Instagram)



Time-Sliced System

- A **Time-Sliced System**, also known as a **Time-Sharing System**, is an operating system that divides the CPU time into small units called **time slices (or time quanta)**.
- Each active process gets a fixed time slice to execute before switching to the next process. This ensures that multiple users or applications can share the CPU fairly.



Multi-processor system:

- A **multi-processor system** is a computer system that uses two or more central processing units (CPUs) within a single machine to improve performance, reliability, and efficiency.
- These processors share memory, system buses, and input/output (I/O) devices, enabling parallel processing.

Types of Multi-Processor Systems

- **Symmetric Multiprocessing (SMP)** – All processors share a single OS and memory, working equally.
- **Asymmetric Multiprocessing (AMP)** – One processor is the master, and others perform assigned tasks.

Advantages: Multi-processor system:

- **Increased Performance** – Tasks are executed in parallel, reducing processing time.
- **Higher Reliability** – If one processor fails, others continue working.
- **Better Resource Utilization** – Efficient workload distribution prevents bottlenecks.

Disadvantages:

- It requires expensive hardware to set up a mainframe computer
- When multiple users log on or work on the same system it reduces the overall performance of the system.

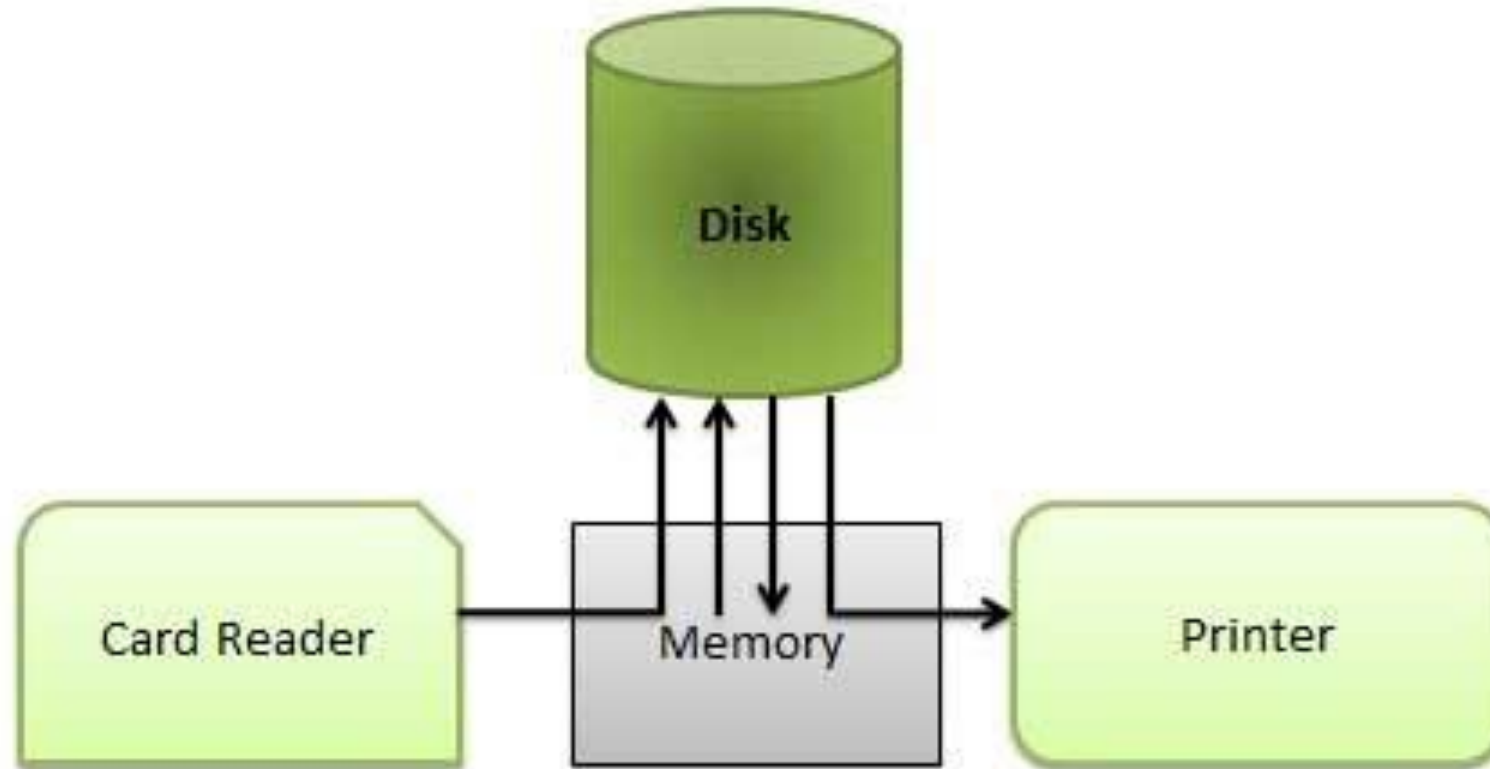
Spooling and Buffer

- Spooling is an acronym for simultaneous peripheral operations on line.
- Spooling refers to putting data of various I/O jobs in a buffer.
- This buffer is a special area in memory or hard disk which is accessible to I/O devices.

An operating system does the following activities

- Maintains the spooling buffer which provides a waiting station where data can rest while the slower device catches up.
- Maintains parallel computation because of spooling process as a computer can perform I/O in parallel fashion. It becomes possible to have the computer read data from a tape, write data to disk and to write out to a tape printer while it is doing its computing task.

Spooling and Buffer



Thank You
Any Questions?